

Advait Rajesh Bharadwaj (he/him)

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EDUCATION

University of Massachusetts Amherst

Amherst, MA, USA

Manning College of Information & Computer Sciences and College of Natural Sciences

GPA: 3.681/4.00

Bachelor of Science in Computer Science and Mathematics (Statistics and Data Science)

Exp. Graduation, May 2026

- **Honors:** Dean's list, \$46,000 Chancellor's Award, **HackUMass XII 2024**, \$7000 Executive Pitch Award, **HackHer(413) 2025**
- **Courses:** Data Structures and Algorithms, Computer Science Principles, Reasoning Under Uncertainty, Discrete Mathematics, Multivariate Calculus, Linear Algebra, Statistics, Artificial Intelligence, Machine Learning

EXPERIENCE

University of Massachusetts Amherst

Amherst, MA

Undergraduate Research Volunteer

May 2024 — September 2024

- Engaged in a research project that involved evaluating the effectiveness of **Classical versus Quantum Machine Learning** models
- Compared these models for performance in **genomic sequence classification** using a dataset with **over 100,000 sequences**
- Analyzed data and applied various machine-learning algorithms to process and classify the genomic sequences
- Demonstrated that classical algorithms, in their current state, consistently outperformed quantum counterparts, providing key insights for future research

Undergraduate Course Assistant for CS383: Artificial Intelligence

September 2024 — Present

- Supported a course with **over 300 students** to enhance their understanding of Artificial Intelligence
- Responded to student inquiries and provided personalized feedback on assignments to improve student performance
- Played a key role in fostering an interactive learning environment, contributing to improved student engagement and academic success

Undergraduate Teaching Assistant for MATH127: Calculus I

September 2023 — Present

- Involved in assisting a large class of **over 1600 students** in mastering challenging mathematical concepts
- Supported lecture delivery, addressed student queries, and **conducted office hours** to ensure smooth course operations
- Organized study groups, helped clarify their doubts, and collaborated closely with professors to manage the class effectively
- Improved the overall class performance by providing timely support and creating a collaborative learning atmosphere

PROJECTS

Roommatez (Flutter, Dart, Firebase)

github/ShadowKrius/Roommatez

Creator and Co-founder

- Developed a cross-platform app matching college roommates based on lifestyle habits
- Built real-time chat, user authentication, and preference-based filtering for compatibility
- Optimized system architecture, caching, and deployment on iOS App Store and Google Play Store
- Utilized agile development methodologies, CI/CD pipeline, and version control with git, and conducted code reviews
- Won **\$7,000** in the Executive Pitch Competition hosted by the Berthiaume Center for Entrepreneurship

SafeSteps (Flutter; Dart, Firebase, Google Maps API)

github/ShadowKrius/safesteps

Awarded **Best UMass Hack** at HackUMass XII 2024

- Developed a campus safety cross-platform app with real-time HELP phone navigation, live location sharing, and emergency alerts
- Integrated Google Maps API for real-time mapping, navigation, and turn-by-turn directions to the nearest HELP phone
- Optimized performance through efficient state management using GetX, enabling smooth and responsive user interactions

CodeStrike Go (React JS; WebSockets, Firebase, CodeMirror, Docker)

github/ShadowKrius/CodeStrike

- Developed a competitive coding platform enabling 1v1 battles with real-time collaborative coding and evaluation
- Implemented matchmaking, scoring, and ELO rating systems for skill-based and balanced competition
- Designed scalable infrastructure using Docker and Firebase to support seamless gameplay and dynamic updates
- Integrated CodeMirror for an interactive code editor and optimized real-time communication with WebSockets

Resonance (Python; Hugging Face; OpenAI API; Google Collab; Librosa API)

github/ShadowKrius/Resonance

Awarded **Best Generative AI Hack – HackHer(413), sponsored by MLH**

- Developed an AI-powered tool that generates dynamic music videos by analyzing audio and lyrics
- Engineered a pipeline using Python, Librosa, and OpenAI API to extract tempo, pitch, and emotion from MP3 files
- Designed a prompt-generation system that converts song characteristics into structured visual descriptions
- Integrated OpenCV and FFmpeg to assemble frames into seamless, synchronized music videos
- Optimized model efficiency and video transitions, ensuring smooth and emotionally resonant outputs

TECHNICAL SKILLS

Languages: Python; Java; C; JavaScript; TypeScript; HTML/CSS; Dart;

Libraries: Pandas; TensorFlow; NumPy; Matplotlib; Scikit-learn; Qiskit;

Tools & Frameworks: Flutter; Firebase; Git; Jupyter; Swift; ReactJS; NextJS;